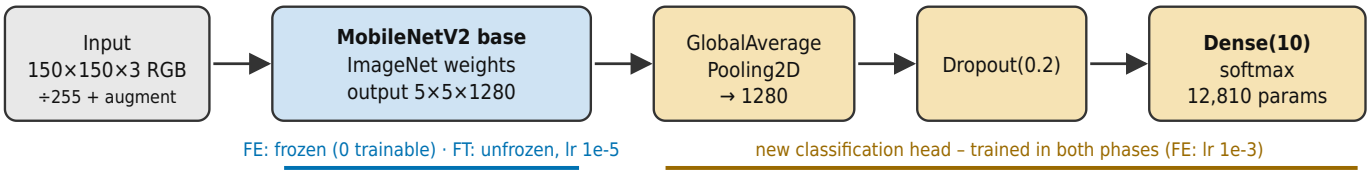
Lab 5 – Transfer Learning: Image Classification Report

Hieu · COS30082 Applied Machine Learning · Dataset: 5,036 labelled images, 10 classes (Animalia, Amphibia, Arachnida, Aves, Fungi, Insecta, Mammalia, Mollusca, Plantae, Reptilia) · split 80/20 → 4,029 train / 1,007 validation · Google Colab T4 GPU

1. Pre-trained model used

MobileNetV2 (Keras Applications, ImageNet weights, `include_top=False`, 2.26 M parameters) is the main model. The same head was also tried on **EfficientNetB0** (4.05 M) and **ConvNeXtTiny** (27.8 M) for comparison. Input 150×150×3, pixels ÷255, augmentation (random flip, rotation 0.1, zoom 0.1) on the training set only; Adam, sparse categorical cross-entropy, EarlyStopping on `val_accuracy` (best weights restored).

2. Model architecture

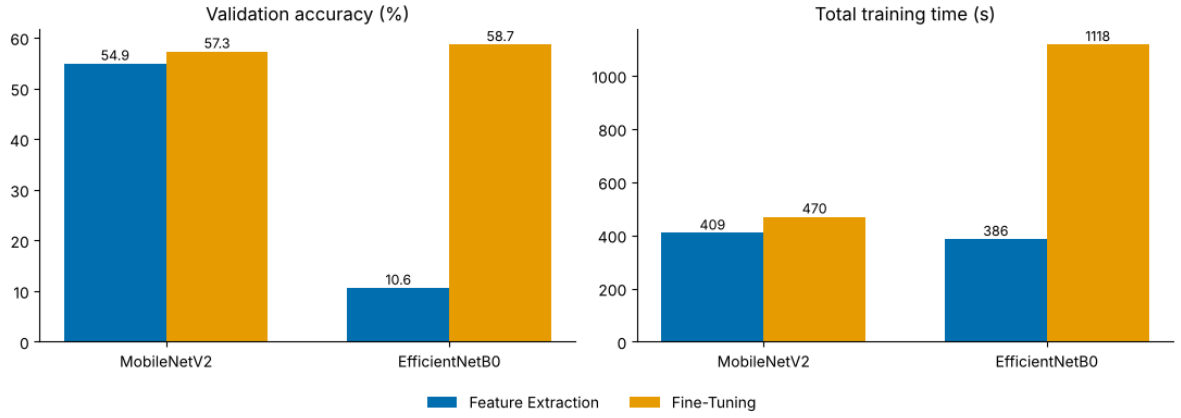


Total 2,270,794 params · FE trainable 12,810 · FT trainable 2,236,682

3. Feature extraction vs fine-tuning

Model	Approach	Trainable params	Epochs	Train acc	Val acc	Train time	s / epoch
MobileNetV2	Feature extraction	12,810	23	61.3%	54.9%	409 s	17.8
MobileNetV2	**Fine-tuning**	**2,236,682**	**22**	**71.6%**	**57.3%**	**470 s**	**21.4**
EfficientNetB0	Feature extraction	12,810	18	10.9%	10.6%	386 s	21.4
EfficientNetB0	**Fine-tuning**	**4,020,358**	**52**	**74.7%**	**58.7%**	**1,118 s**	**21.5**
ConvNeXtTiny	Feature extraction	7,690	16	27.7%	30.8%	340 s	21.2
ConvNeXtTiny	Fine-tuning	not completed (runtime reset)					

Accuracy = best validation accuracy (train accuracy at the same epoch, on augmented data). Time = sum of Keras epoch times until early stopping.



4. Analysis

- **Accuracy:** fine-tuning is better for every model that completed it. On MobileNetV2 – the cleanest comparison – it raises validation accuracy from 54.9% to 57.3% (+2.4 pts); train accuracy rises much more (61% → 72%), so the train/validation gap widens: the 4k-image set is small and fine-tuning overfits.
- **Training time:** feature extraction is faster per epoch (17.8 s vs 21.4 s, no gradients through the base) and the whole run is ~13% shorter (409 s vs 470 s) for MobileNetV2. EfficientNetB0 fine-tuning needed 52 epochs (1,118 s) to converge.
- **Trade-off:** fine-tuning gives a modest gain for slightly more time and a higher overfitting risk; feature extraction is the cheaper, safer baseline.
- **Limitations:** (i) EfficientNetB0 / ConvNeXtTiny already contain their own preprocessing but received ÷255 inputs, which most likely explains the 10.6% / 30.8% feature-extraction results, so their FE-vs-FT gap is exaggerated; (ii) fine-tuning unfroze the whole base model with lr 1e-5, not only the top layers; (iii) ConvNeXtTiny fine-tuning was not completed; (iv) single run per setting, validation set only (the test folder is unlabelled).